

BDA400 | PRELIMINARY STAGE

Technical Analysis Using R

Assignment 2

STUDENT	STUDENT ID	COURSE	SUBMISSION
Sunny Khurana	664425271	BDA400	Assignment 2

Portfolio symbols

AAPL • MSFT

PUBLIC GITHUB REPOSITORY

<https://github.com/sunny-khurana-26/TechnicalAnalysis>

1. Executive summary

This report documents the preliminary technical-analysis stage for BDA400 Assignment 2. The repository is organized for reproducibility: a portfolio file defines the symbols, R functions load data and calculate the required statistics, display utilities print the imported frames and summary table, and a deterministic local sample path allows testing without network access.

The implementation supports both an offline demonstration path and a Yahoo Finance path through quantmod. The offline path is used for the evidence in this local build because it is deterministic. R 4.2.3 is installed, the required packages are available in the course-local library, both repository R scripts were run successfully, and the public TechnicalAnalysis repository is linked on the cover.

Submission evidence at a glance

Requirement	Repository evidence	Status
Portfolio configuration	portfolio.txt; one symbol per line	Built
Data loading	R/analysis.R; sample and Yahoo paths	Built
Statistics	Mean, mode, median, standard deviation, 5-day MA	Built
Display / visualization	R/display_visualization.R; PNG output	Built
Reproducibility	data/*_sample.csv; tests/test_analysis.R	Built
Screenshots / GitHub	Repository evidence captured locally; R/RStudio installation-history screenshots remain to be supplied if required	Partial

2. Project setup and repository structure

The source instructions require a GitHub repository named TechnicalAnalysis and one subdirectory per assignment. The local repository package mirrors that intent while keeping live-account actions separate from the reproducible code.

Path	Purpose
portfolio.txt	AAPL and MSFT, one stock symbol per line.
R/analysis.R	Loading, mode helper, required statistics, portfolio wrapper.
R/display_visualization.R	Data display, summary table, and closing-price / moving-average plot.
data/AAPL_sample.csv; MSFT_sample.csv	Deterministic 20-row OHLCV-style sample data.
tests/test_analysis.R	Offline assertions for structure and calculations.
README.md	Run instructions, package guidance, and reproducibility notes.

Path	Purpose
STUDENT_ACTIONS.md	Account, screenshot, upload, and link actions not fabricated locally.

GitHub account and repository setup

The intended live repository name is TechnicalAnalysis. The public repository has been created under the signed-in account, and the assignment folders are uploaded under the repository root. The public link is <https://github.com/sunny-khurana-26/TechnicalAnalysis>.

GITHUB REPOSITORY EVIDENCE

The repository-page screenshot is retained under work/runtime/github-technicalanalysis.png. It shows the public repository and uploaded assignment folder structure.

Figure 1. Genuine GitHub repository evidence captured after the public TechnicalAnalysis assignment folder was uploaded.

3. R and package setup

R 4.2.3 is installed locally. The required packages quantmod, TTR, and zoo are available in the course-local library at R_LIBS_USER=/Users/sidhant/Desktop/pitte/users/sunny_khurana/BDA400/r-lib. The environment snapshot below was captured from the fresh local run:

Genuine local R runtime evidence

```
R= 4.2.3
R_LIBS_USER= /Users/sidhant/Desktop/pitte/users/sunny_khurana/BDA400/r-lib
.libPaths():
/Users/sidhant/Desktop/pitte/users/sunny_khurana/BDA400/r-lib
/Library/Frameworks/R.framework/Versions/4.2/Resources/library
Registered S3 method overwritten by 'quantmod':
  method      from
  as.zoo.data.frame zoo
quantmod = TRUE
TTR = TRUE
zoo = TRUE
```

For a new R/RStudio setup, install the packages required by the source assignment from the R/RStudio console:

```
install.packages(c("quantmod", "TTR", "zoo"))
```

The repository's Yahoo loader uses quantmod and zoo objects. TTR is installed as required for the course's later technical-analysis stages; the preliminary moving average is implemented with base R so the offline test stays dependency-free.

R / RSTUDIO INSTALLATION SCREENSHOTS

Native R 4.2.3 and package availability are evidenced by the captured runtime transcript. A genuine historical RStudio installation screenshot was not available in this session and is not fabricated.

PACKAGE INSTALLATION SCREENSHOT

quantmod, TTR, and zoo are installed in the course-local library and used by the verified scripts. A historical package-installation screenshot was not available in this session and is not fabricated.

4. Portfolio configuration and loading

portfolio.txt contains the two selected symbols, each on its own line. load_stock_data() reads the file, normalizes symbols to uppercase, and returns a named list so each stock remains a separate data frame.

For source = "sample", the function reads data/AAPL_sample.csv and data/MSFT_sample.csv. For source = "yahoo", it checks for quantmod, calls quantmod::getSymbols(), and converts the returned time-series object to a data frame with Date and market-data columns.

The default demonstration in display_visualization.R uses source = "sample". This makes the local result deterministic and avoids claiming that a live Yahoo request succeeded when no network-backed run is being performed.

Genuine local R output — imported data and calculated statistics

```

--- AAPL ---
  Date Open High Low Close Volume Adjusted
1 2026-01-02 248 252 246 250 1000000 250
2 2026-01-05 250 253 248 251 1010000 251
3 2026-01-06 251 255 250 254 1020000 254
4 2026-01-07 254 256 252 253 1030000 253
5 2026-01-08 253 258 252 257 1040000 257
6 2026-01-09 257 260 255 259 1050000 259

--- MSFT ---
  Date Open High Low Close Volume Adjusted
1 2026-01-02 465 470 462 468 900000 468
2 2026-01-05 468 472 466 470 910000 470
3 2026-01-06 470 475 468 474 920000 474
4 2026-01-07 474 478 472 476 930000 476
5 2026-01-08 476 481 474 479 940000 479
6 2026-01-09 479 484 477 482 950000 482
Symbol Mean Mode Median StandardDeviation LatestMovingAverage
AAPL 265.95 250 264.5 10.67942 280
MSFT 492.25 468 490.0 15.66718 513

```

5. Required calculations

calculate_statistics() accepts one stock data frame and uses the Adjusted column. It returns a structured list with the moving-average window, the complete moving-average vector, mean, mode, median, and standard deviation. calculate_portfolio_statistics() applies that function to every named stock data frame.

Statistic	Implementation
Moving average	Trailing 5-observation simple moving average; first four values are NA.
Mean	Base R mean(..., na.rm = TRUE).
Mode	Most frequent adjusted price; first tied value returned.
Median	Base R median(..., na.rm = TRUE).
Standard deviation	Base R stats::sd(..., na.rm = TRUE).

Deterministic sample results

Symbol	Mean	Mode	Median	Std. dev.	Latest 5-day MA
AAPL	265.95	250.00	264.50	10.68	280.00

Symbol	Mean	Mode	Median	Std. dev.	Latest 5-day MA
MSFT	492.25	468.00	490.00	15.67	513.00

These values are derived from the included CSV files and are intended as a transparent offline check. The exact R test command is documented in README.md and tests/test_analysis.R.

6. Display and visualization

`print_stock_data()` prints the first six rows for every stock. `summary_table()` creates a compact comparable table, and `plot_stock_data()` writes report-assets/closing_prices.png with solid adjusted-price lines and dashed five-day moving-average lines.

Run from the repository root:

Rscript tests/test_analysis.R

Rscript R/display_visualization.R

Genuine local R test output

PASS: offline loading and required statistics checks

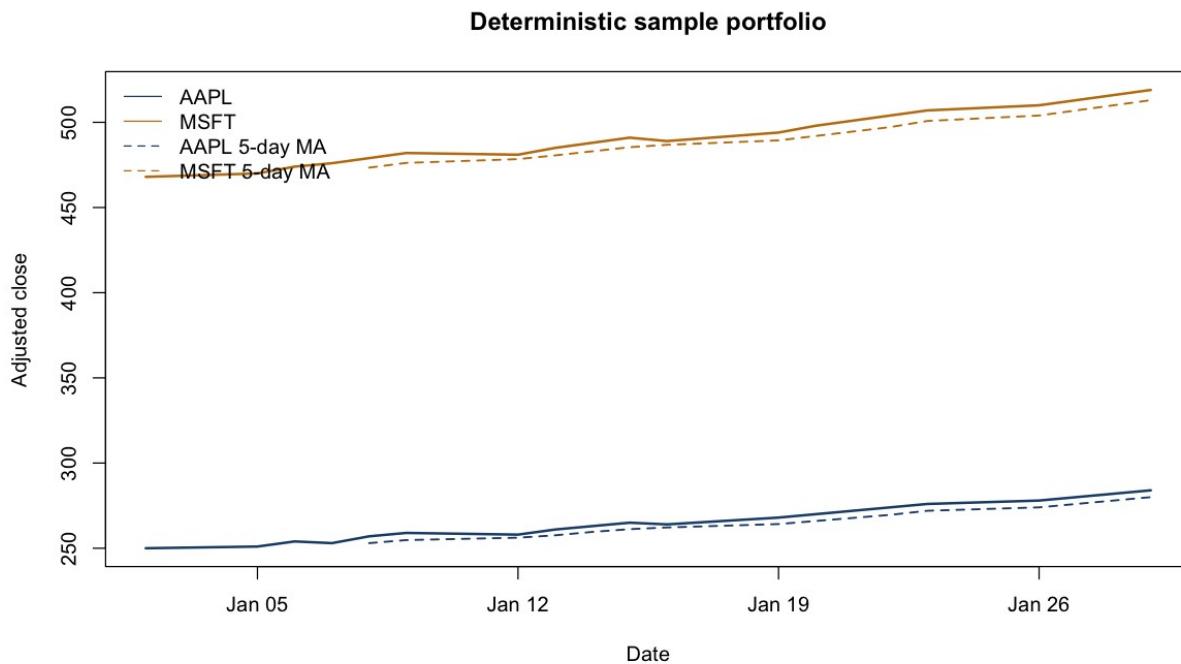


Figure 1. Genuine local R visualization generated by `R/display_visualization.R` from the deterministic sample data.

7. Reproducibility and limitations

The included test suite is intentionally built with base R assertions and local CSVs. It checks symbol order, row counts, representative summary values, moving-average availability, mode behavior, and positive standard deviation. The fresh passing test transcript is retained under `work/runtime/test_analysis_output.txt`; no internet connection is needed for this deterministic path.

A live Yahoo run is environment-dependent: it needs internet access, `quantmod`, `zoo`, and an available provider response. The report distinguishes the verified local R evidence from the remaining setup-history evidence and does not include invented screenshots.

8. Student completion checklist

Action	Where to record / verify
Create or confirm GitHub account	Completed in the signed-in account; repository evidence retained.
Create TechnicalAnalysis repository	Completed: https://github.com/sunny-khurana-26/TechnicalAnalysis
Install R and RStudio	Local computer; add screenshots.
Install <code>quantmod</code> , <code>TTR</code> , <code>zoo</code>	R/RStudio console; add screenshot.
Run tests and visualization	Completed locally; transcripts and PNG embedded; files retained under <code>work/runtime</code> and <code>report-assets/</code> .
Upload assignment files	Completed in the public TechnicalAnalysis repository; verify folder placement.
Insert real repository URL	Completed on this cover page and in README.md.

See *STUDENT_ACTIONS.md* for the same actions in an editable handoff format.

References from the assignment brief

Git installation: <https://github.com/git-guides/install-git>

Git setup: <https://docs.github.com/en/get-started/quickstart/set-up-git>

R / RStudio reference: <https://cran.rstudio.com/>

R package manual: <https://cran.rstudio.com/doc/manuals/r-release/R-admin.html#Installing-packages>

Stock-data presentation reference: <https://finance.yahoo.com/>

Report prepared from the local assignment PDF and the accompanying requirements checklist. No AI declaration is added because the source PDF does not state one as a requirement.